

Statistical Evaluation of Delay and Doppler Spreads in sub-6 GHz and mmWave Vehicular Channels

F. Pasic, M. Hofer, M. Mussbah, H. Groll, T. Zemen,
S. Schwarz and C. F. Mecklenbräuer

Presenter: Faruk Pasic

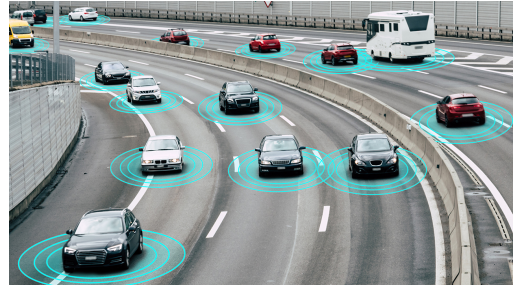


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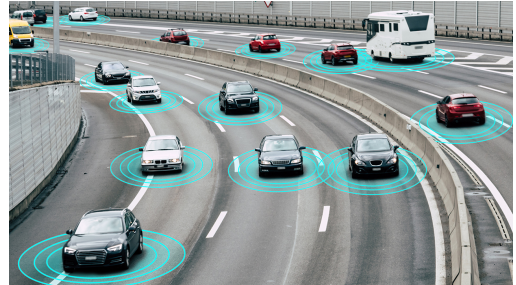
Introduction

- increasing demand for high data rate transmission



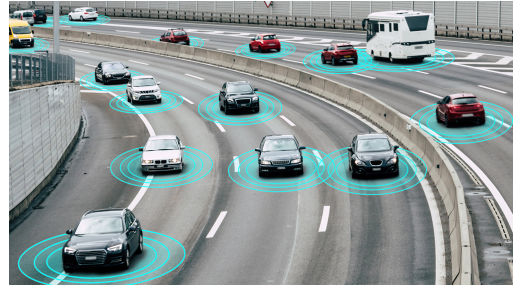
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- increasing demand for high data rate transmission
 - high-mobility environments



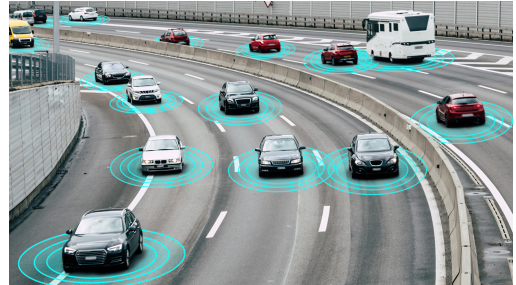
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- increasing demand for high data rate transmission
 - high-mobility environments
 - sub-6 GHz band cannot meet these requirements



Introduction

- increasing demand for high data rate transmission
 - high-mobility environments
 - sub-6 GHz band cannot meet these requirements
- mmWave bands allow for significantly higher data rates



Motivation

- understand differences and similarities in terms of propagation channel characteristics

[1] D. Dupleich et al., “Multi-band propagation and radio channel characterization in street canyon scenarios for 5G and beyond,” (IEEE Access 2019)

[2] M. Hofer et al., “Wireless Vehicular Multiband Measurements in Centimeterwave and Millimeterwave Bands” in International Symposium on Personal, Indoor and Mobile Radio Communications (PIMRC 2021)

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- only few comparative multi-band analyses in the literature

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 - statical measurements [1]: only frequency selectivity analysis
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 - problem of antenna placement

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- **multi-band high-mobility wireless channel measurements conducted**
 - identical transmit and receive antenna positions
 - modified version of the measurement setup proposed in [3]
 - methodology proposed in [4]

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- **multi-band high-mobility wireless channel measurements conducted**
 - identical transmit and receive antenna positions
 - modified version of the measurement setup proposed in [3]
 - methodology proposed in [4]
- **comparative study of the multi-band propagation based on measurement results**
 - estimate the power delay profile (PDP) and the Doppler power spectral density (DSD)
 - compare channels in terms of RMS delay spread and RMS Doppler spread

[3] F. Pasic et al. "Sub 6 GHz versus mmwave Measurements in a Controlled High-mobility Environment" in 25th International ITG Workshop on Smart Antennas (WSA 2021)

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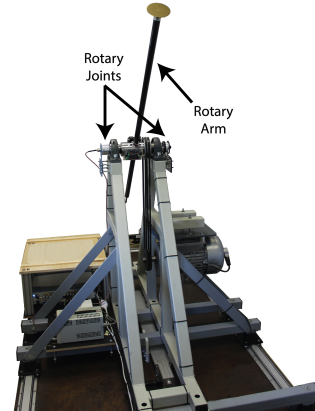
Measurement Campaign

Results and Discussion

Summary and Conclusion

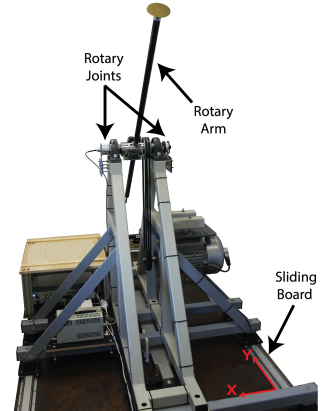
Measurement Setup

- moving transmitter and static receiver
 - rotary unit
 - trigger unit



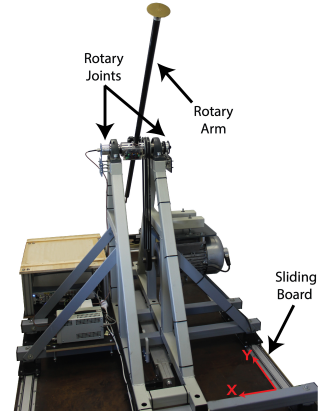
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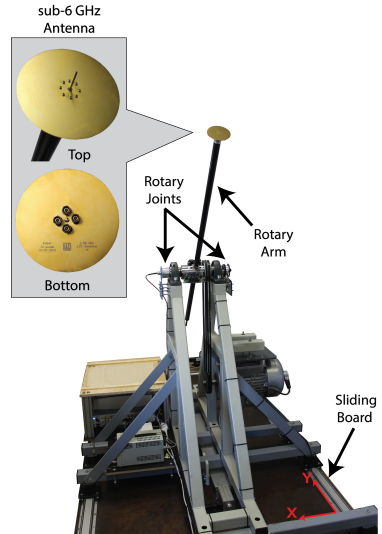
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- operates at 2.55 GHz and 25.5 GHz



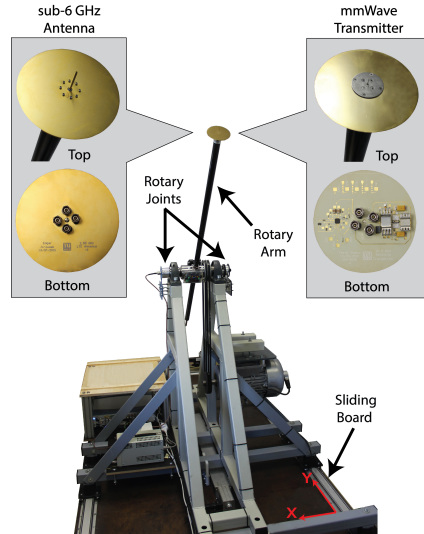
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 - sub-6 GHz: direct transmission



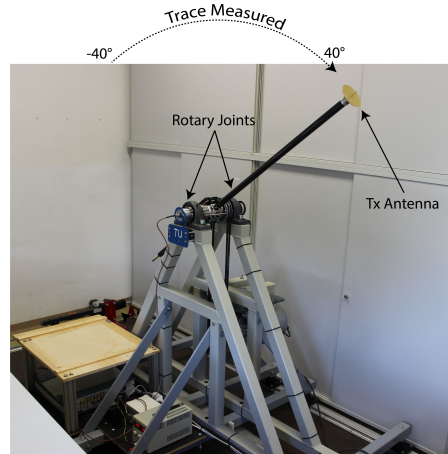
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 - mmWave: lightweight transmitter



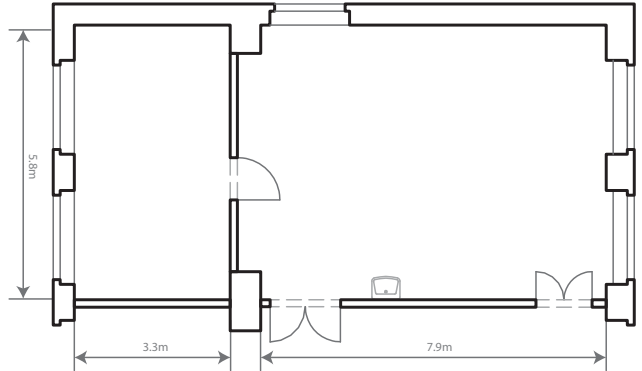
Measurement Setup

- moving transmitter and static receiver
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 - sliding board
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 - sub-6 GHz: direct transmission
 - mmWave: lightweight transmitter
- perform wireless channel measurements along the same trace for different frequency bands



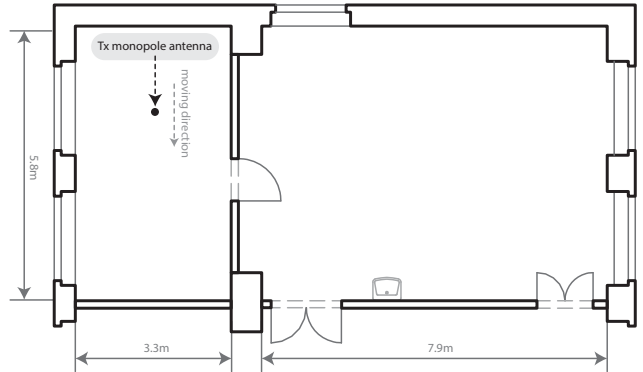
Measurement Procedure

- wireless channel measurements



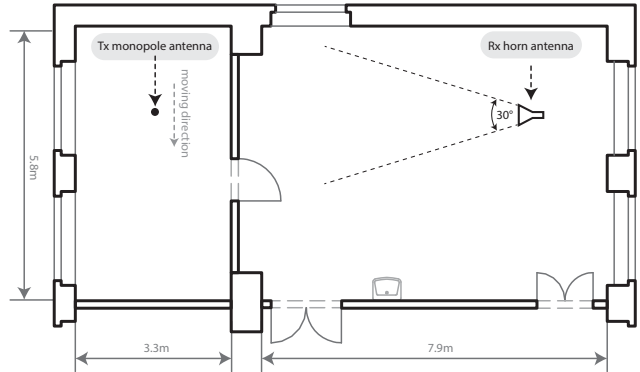
Measurement Procedure

- wireless channel measurements
 - moving Tx antenna



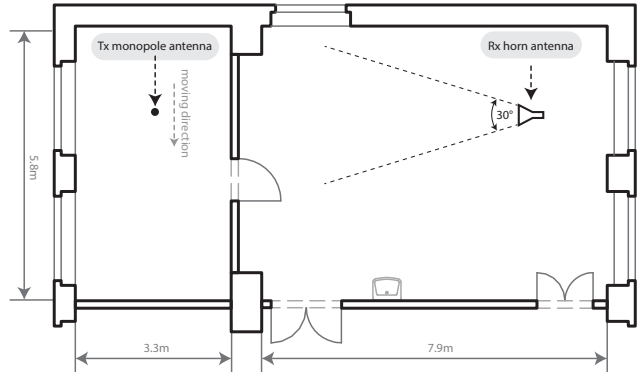
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Measurement Procedure

- wireless channel measurements
 - moving Tx antenna
 - static Rx antenna
 - static fading environment



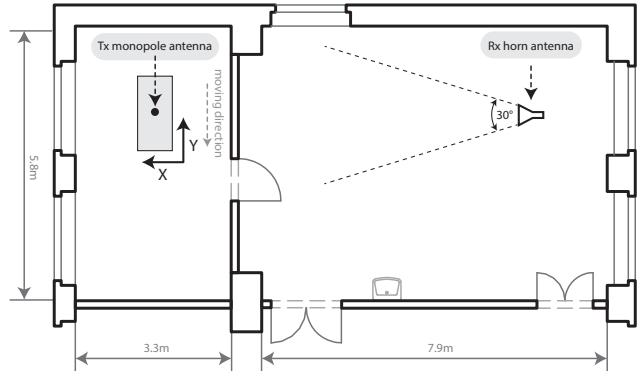
Measurement Procedure

- wireless channel measurements
 - moving Tx antenna
 - static Rx antenna
 - static fading environment
- measurement procedure
 - channel sounding using OFDM
 - center frequency: 2.55 GHz or 25.5 GHz
 - velocity: 50 km/h or 100 km/h

Parameter	Value			
Bandwidth B [MHz]	100			
Delay Resolution $\Delta\tau$ [ns]	10			
Number of Snapshots N	500			
Symbols per Snapshot N_{sym}	100			
Number of x-axis Positions N_x	7			
Number of y-axis Positions N_y	18			
Measured Arc Segment	$-40^\circ \dots 40^\circ$			
	A	B	C	D
Carrier Frequency f_c [GHz]	2.55	2.55	25.5	25.5
Wavelength λ [cm]	11.7	11.7	1.17	1.17
Transmitter Velocity v_{Tx} [km/h]	50	100	50	100
Number of Subcarriers K	200	100	200	100
Subcarrier Spacing Δf [kHz]	500	1000	500	1000
Max. Doppler Shift ν_{max} [Hz]	118.1	236.2	1181	2362

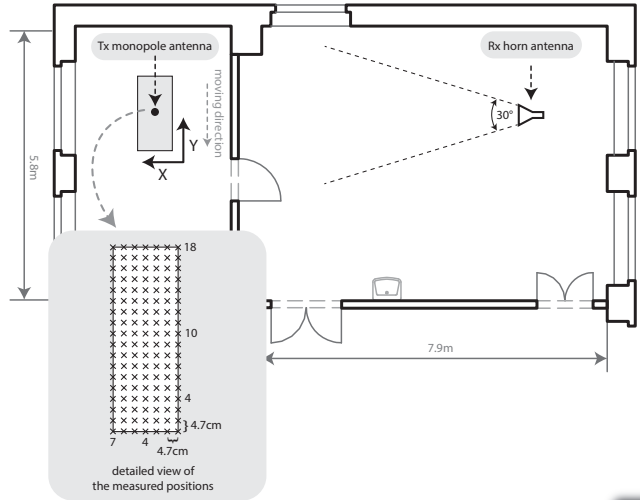
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 - center frequency: 2.55 GHz or 25.5 GHz
 - velocity: 50 km/h or 100 km/h
 - 126 different spatial positions



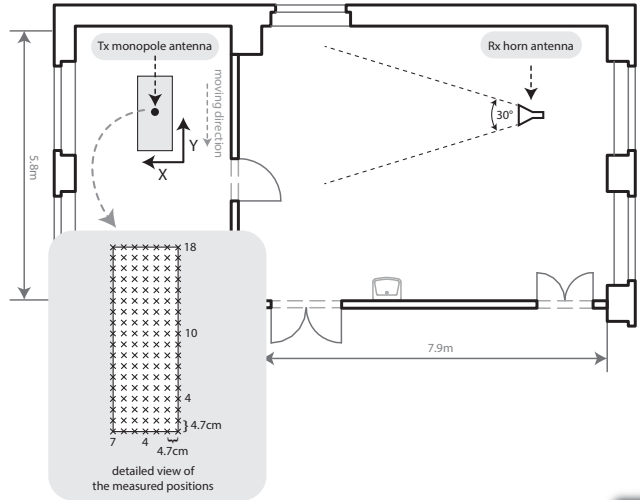
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- result
 - time-variant channel transfer function
 - RMS delay spread
 - normalized RMS Doppler spread



Contents

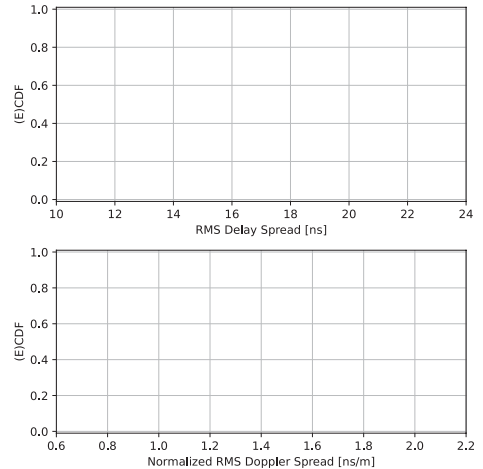
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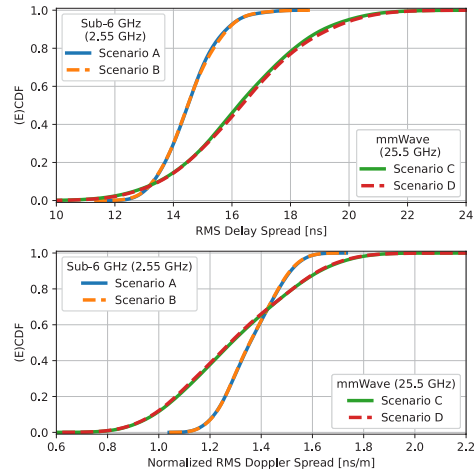
Results and Discussion

- cumulative distribution function (CDF)



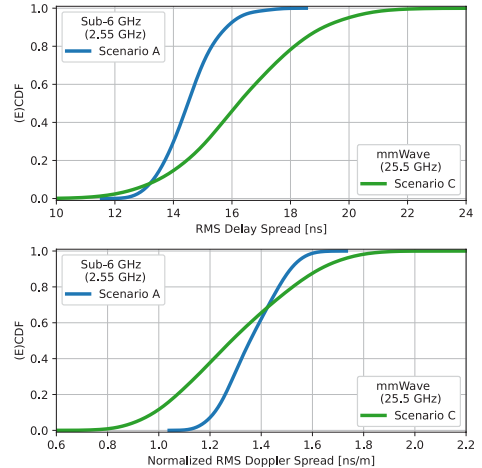
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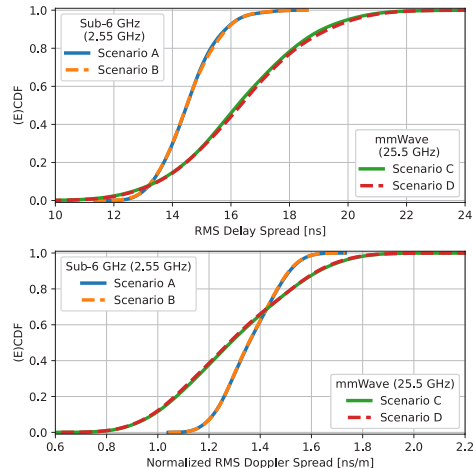
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- cumulative distribution function (CDF)
- invariance with regard to the transmitter's velocity



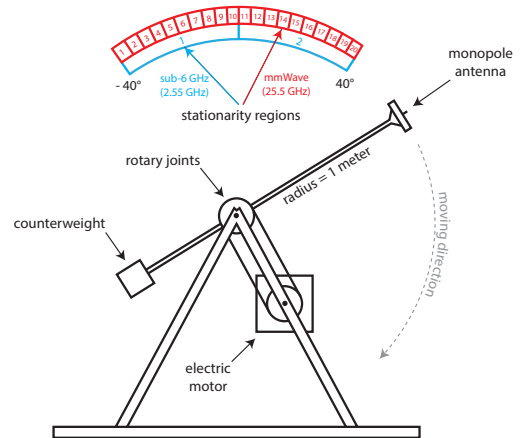
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- cumulative distribution function (CDF)
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- higher standard deviation for mmWave



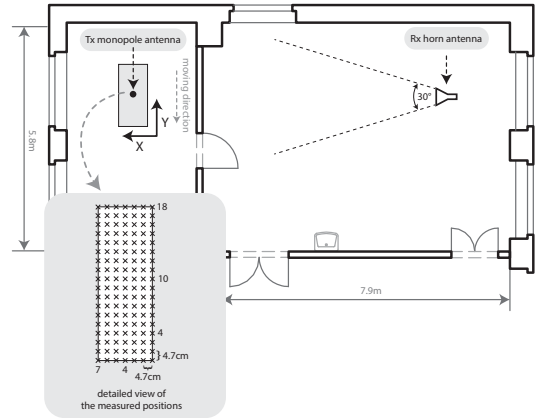
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 - more mmWave stationarity regions



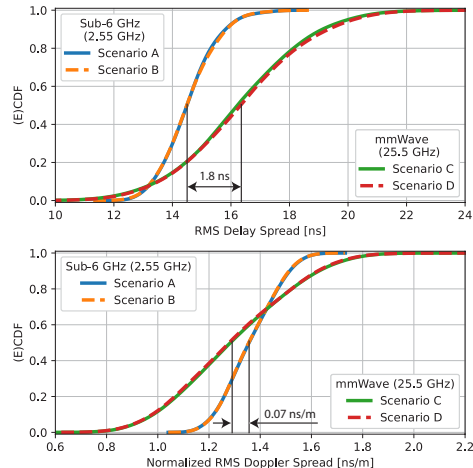
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- cumulative distribution function (CDF)
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 - more mmWave stationarity regions
 - distance of 4λ between spatial positions



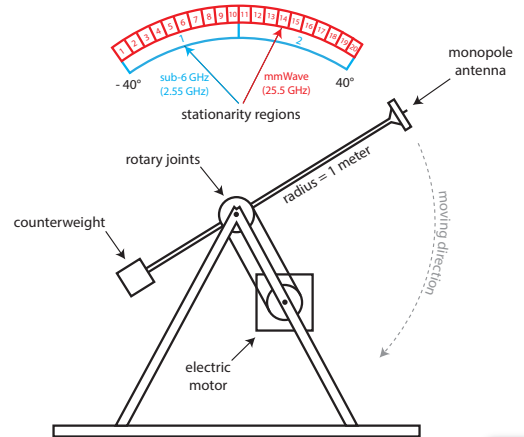
Results and Discussion

- cumulative distribution function (CDF)
- invariance with regard to the transmitter's velocity
- higher standard deviation for mmWave
 - more mmWave stationarity regions
 - distance of 4λ between spatial positions
- mean values very similar
 - RMS delay spread: 12% difference
 - normalized RMS Doppler spread: 5% difference



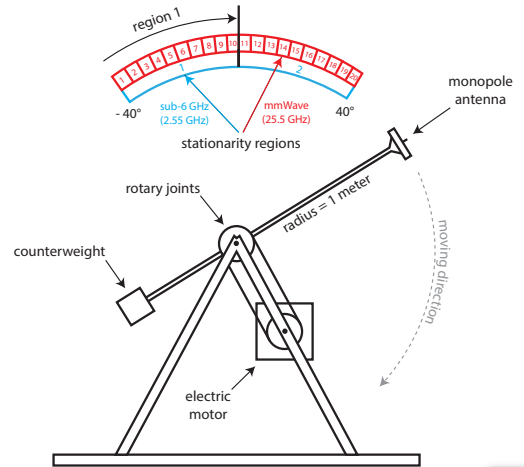
Results and Discussion

- clustering into 2 regions



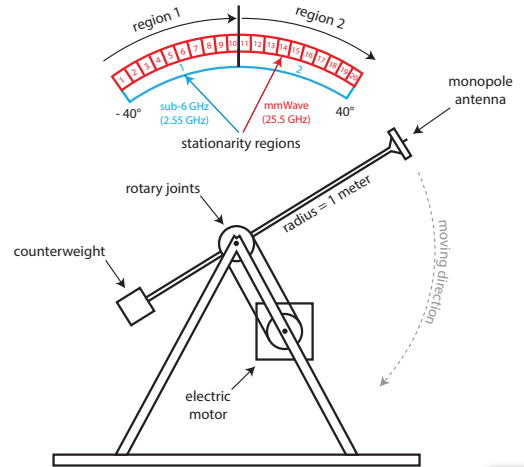
Results and Discussion

- clustering into 2 regions
 - region 1



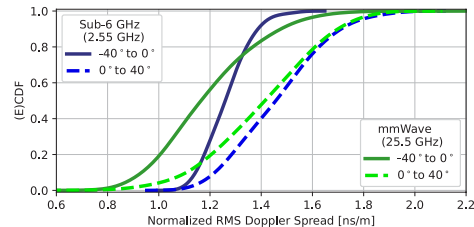
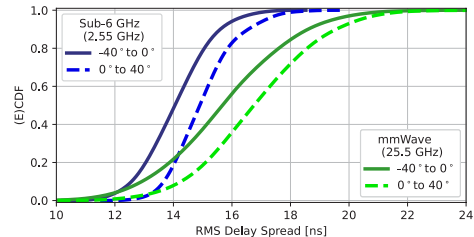
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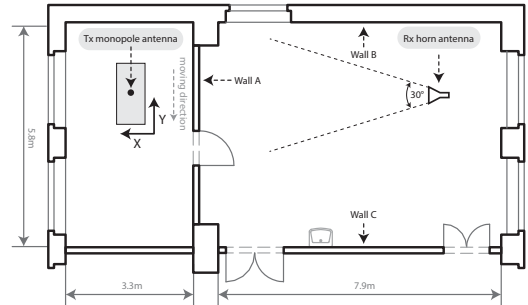
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 - cumulative distribution function (CDF)



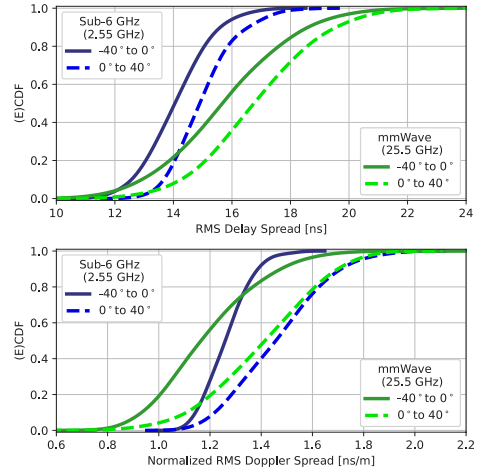
Results and Discussion

- clustering into 2 regions
 - region 1
 - region 2
 - cumulative distribution function (CDF)
- region 1
 - blockage from wall A
 - reflections from wall B



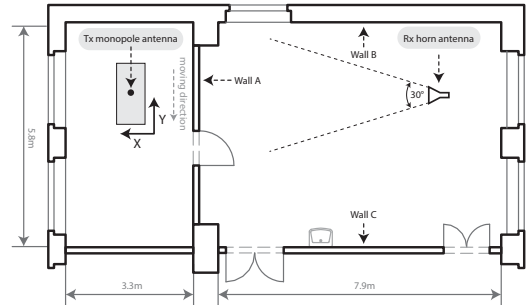
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 - blockage from wall A
 - reflections from wall B
 - lower values of channel spreads



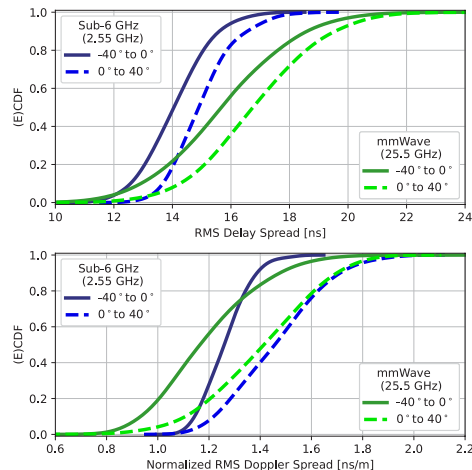
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 - reflections from wall B
 - lower values of channel spreads
- region 2
 - many unblocked components pass into the neighboring room



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 - region 1
 - region 2
 - cumulative distribution function (CDF)
- region 1
 - blockage from wall A
 - reflections from wall B
 - lower values of channel spreads
- region 2
 - many unblocked components pass into the neighboring room
 - channel spreads larger by 7 % and 16 %



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- channel spreads invariant with regard to the transmitter's velocity
 - **measurement setup to compare sub 6 GHz and mmWave bands works correctly**

Summary and Conclusion

- channel spreads invariant with regard to the transmitter's velocity
 - **measurement setup to compare sub 6 GHz and mmWave bands works correctly**
- measured results show small differences on the channel spreads
 - **propagation conditions at different frequency bands are very similar**

Thank You !

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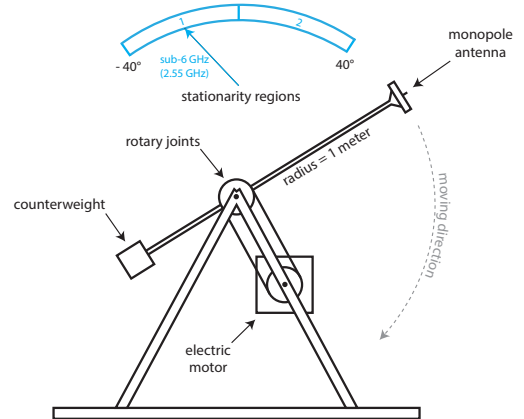
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- estimate local scattering function

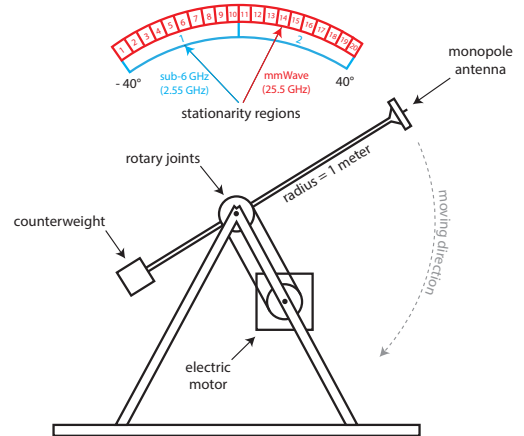
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 - 6λ of motion



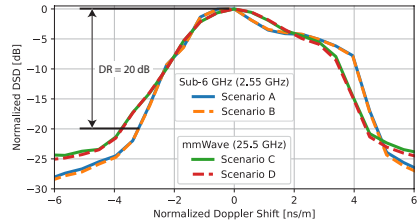
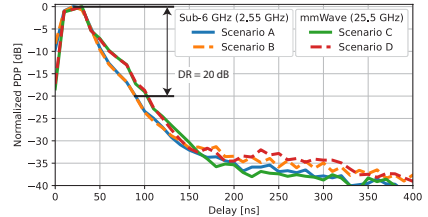
Measurement Evaluation

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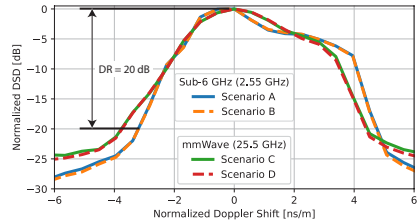
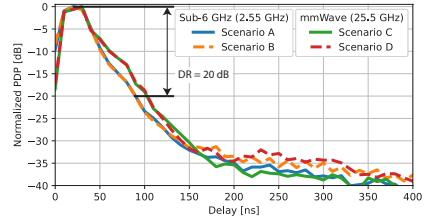
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 - entire frequency range
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 - calculate the PDP and DSD



Measurement Evaluation

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 - entire frequency range
 - 6λ of motion
 - calculate the PDP and DSD
- adjustments performed:
 - power of PDP and DSD normalized to 0 dB
 - define dynamic range (DR)
 - normalized Doppler Shift $\nu_N = \frac{\nu}{f_c v_{Tx}}$



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- adjustments performed:
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- result: RMS delay spread and normalized RMS Doppler spread

